

# Web Technologies Frontend

CASCADING STYLE SHEETS - CSS

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# CSS

Cascading Style Sheets, fondly referred to as CSS, is a simple-designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

## **CSS Types:**

- Inline CSS
- Internal or Embedded CSS
- External CSS

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# Inline CSS

Inline CSS contains the CSS property in the body section attached to the element, which is known as inline CSS. This kind of style is specified within an HTML tag using the style attribute.

```
<!DOCTYPE html>
<html>
<head>
  <title>Inline CSS</title>
</head>

<body>
<p style = "color:#009900; font-size:50px;
           font-style:italic; text-align:center;">
  Inline CSS Example
</p>
</body>
</html>
```

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# Internal CSS

This can be used when a single HTML document must be styled uniquely. The CSS rule set should be within the HTML file in the head section i.e, the CSS is embedded within the HTML file.

```
<!DOCTYPE html>
<html>
<head>
  <title>Internal CSS</title>
  <style>
    .main {
      text-align:center;
    }
    .class1 {
      color:#009900;
      font-size:50px;
      font-weight:bold;
    }
    .class2 {
      font-style:bold;
      font-size:20px;
    }
  </style>
</head>
```

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# External CSS

External CSS contains a separate CSS file which contains only style properties with the help of tag attributes (For example, class, id, heading, etc). CSS property written in a separate file with a .css extension and should be linked to the HTML document using the link tag

HTML File

```
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" href="ex.css"/>
</head>
```

CSS File

```
body {
    background-color: powderblue;
}
.main {
    text-align: center;
}
```

---

# CSS Selectors

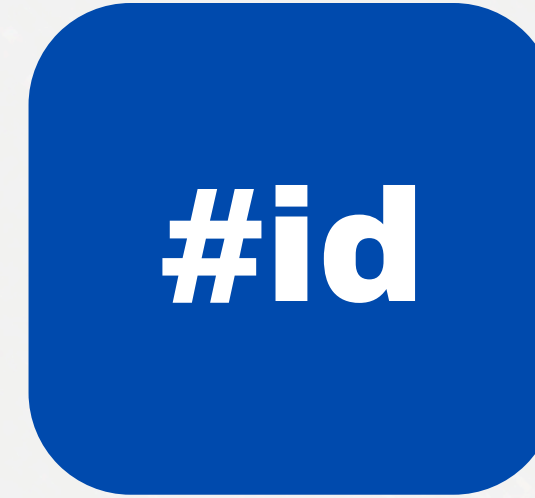
CSS selectors are used to "find" (or select) the HTML elements you want to style.

- Element selectors (e.g., `h1`, `p`) for general styling of HTML elements.
- Class selectors (e.g. `.button-style`) to style elements that share a class.
- ID selectors (e.g., `#header`, `#main`) to target specific, unique elements.
- Descendant selectors (e.g., `.nav a`) to style elements nested within other elements
- Universal selector (`*`) to style all HTML elements on the page



Element

```
p { color: green; }
```



ID

```
#id { color: green; }
```



Class

```
.class { color: green; }
```



Descendant selector

```
nav a { color: green; }
```

# — Specificity

CSS specificity determines which styles are applied when multiple rules target the same element. It's based on the type of selector:

1. **!important** rule forces a style to override any other rules, regardless of specificity.
2. **Inline** styles (e.g., `style="color: red;"`) have the next highest specificity.
3. **IDs** (e.g., `#header`) are more specific than classes or elements.
4. **Classes, pseudo-classes, and attributes** (e.g., `.btn`, `:hover`, `[type="text"]`) have medium specificity.
5. **Elements and pseudo-elements** (e.g., `p`, `::before`) have the lowest specificity.

More specific rules override less specific ones. If specificity is the same, the last declared rule wins.

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# CSS Units

Absolute units (fixed size):

***px (pixels)***

*common for screen design*

Relative units (scale based on context):

***%***

*relative to parent element*

***em***

*relative to parent font size*

***rem***

*relative to root (HTML) font size*

***vw, vh***

*relative to viewport width/height*

# — Text formatting

***color***

*sets the text color, can be specified by color name, HEX or RGB value*

***text-align***

*sets horizontal alignment of the text*

***font-size***

*px, em, rem*

***direction***

*changes text direction*

***decoration***

*adds decoration to the text*

***text-transform***

*changes letter case to uppercase or lowercase*

# Web Safe Fonts

Web-safe fonts are widely supported by most web browsers and operating systems. They don't rely on custom font downloads, making websites faster and more reliable.

## Serif

Times New Roman

Georgia

Cambria

## Sans-Serif

Arial

Calibri

Trebuchet MS

Century Gothic

## Monospace

Courier New

Courier

## Cursive

Comic Sans MS

# — Images

| Method            | Purpose                                          | Code Example                                                                                                                           |
|-------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Background Images | Set an image as the background of an element     | <pre>.element {<br/>  background-image: url('image.jpg');<br/>  background-size: cover;<br/>  background-position: center;<br/>}</pre> |
| Images as Content | Insert an image as part of the element's content | <pre>.icon::before {<br/>  content: url('icon.png');<br/>}</pre>                                                                       |
| Border Images     | Use an image for the border of an element        | <pre>.box {<br/>  border: 10px solid transparent;<br/>  border-image: url('border.png') 30<br/>round;<br/>}</pre>                      |

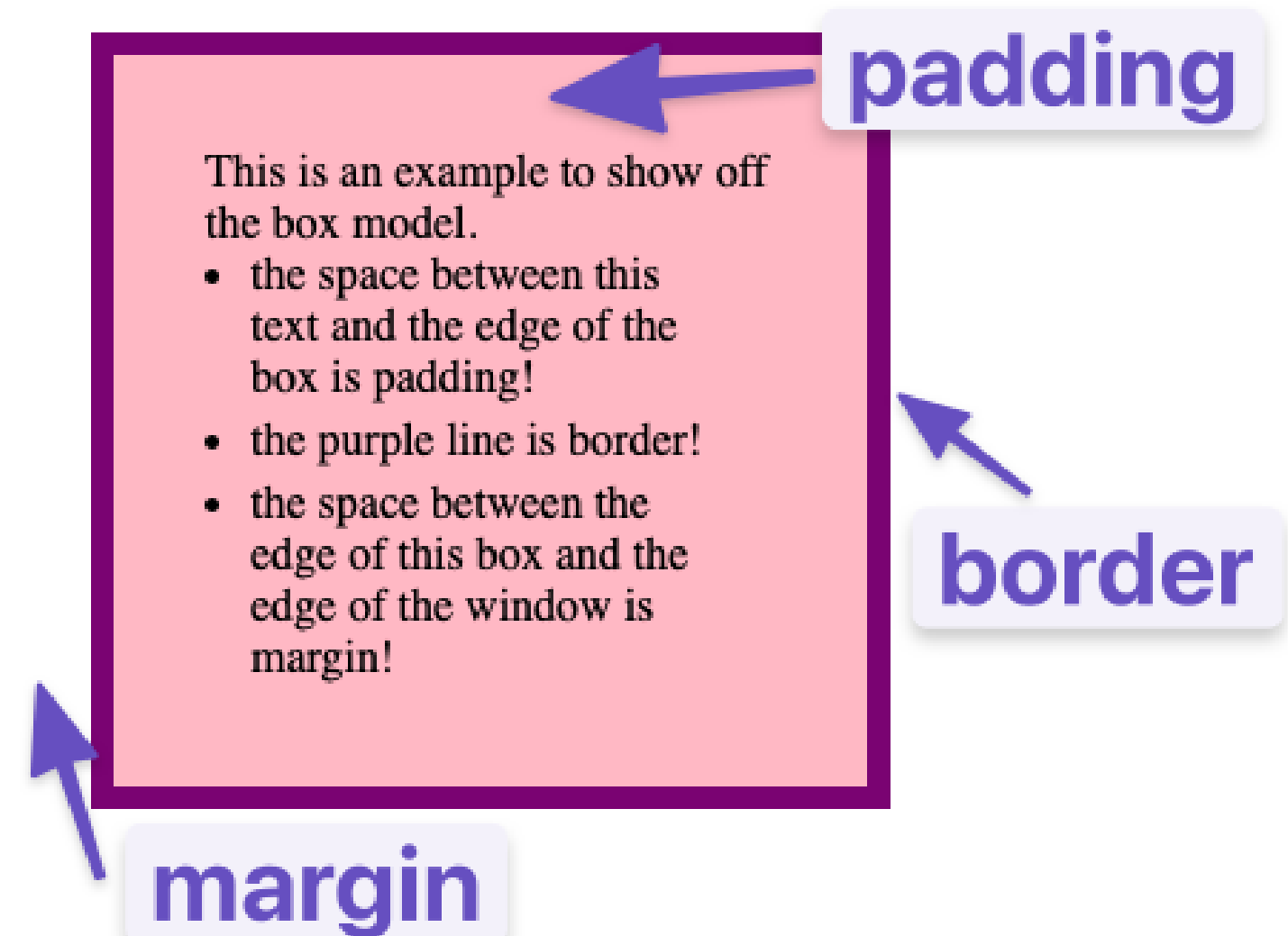
# Box Model

Every element is a box:

- Content - text, images
- Padding - space inside box
- Border - line around content
- Margin - space outside box

We can check how it works by applying an outline to every element on the page:

```
* {  
  outline: 2px solid red;  
}
```



# Positioning

| Property        | Description                                                             | Example Use Cases                                 |
|-----------------|-------------------------------------------------------------------------|---------------------------------------------------|
| <b>static</b>   | Default; follows the normal document flow.                              | No special positioning.                           |
| <b>relative</b> | Positioned relative to its normal position.                             | Positioned 20px from the top, 30px from the left. |
| <b>absolute</b> | Positioned relative to the nearest positioned ancestor or the viewport. | Moves 10px down from its original spot.           |
| <b>fixed</b>    | Stays in the same position on the screen when scrolling.                | Always stays at the bottom-right of the viewport. |
| <b>sticky</b>   | Switches between relative and fixed depending on scroll position.       | Sticks to the top when scrolled to that point.    |

THANK YOU FOR LISTENING

***Any questions?***

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